

PAPER CODE: BEM-C202
SEMESTER EXAMINATION (MAY-2025)
CLASS: B.Tech. SEMESTER: II
SUBJECT: ENGINEERING MATHEMATICS II

Time: 3 Hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A** and **B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)		CO	BL										
Instructions: Answer any <i>five</i> questions in about 150 words each. Each question carries six marks. (5 x 6 = 30 Marks)													
Q-1	Solve $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = \cos x + e^{-2x}$	CO1	L1										
Q-2	Solve $x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} - 12y = x^2 \log x$	CO1	L5										
Q-3	Solve $\frac{\partial^3 u}{\partial x^3} + \frac{\partial^3 u}{\partial x^2 \partial y} + 4 \frac{\partial^3 u}{\partial y^3} = e^{x+2y}$	CO2	L2										
Q-4	Solve $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \sin x$	CO2	L5										
Q-5	Prove that $J_2'(x) = \left(1 - \frac{4}{x^2}\right) J_1(x) + \frac{2}{x} J_0(x)$	CO3	LL3										
Q-6	Express $f(x) = x^3 + 1$ in terms of Legendre Polynomial.	CO3	L4										
Q-7	Find the Fourier series for $f(x) = \frac{1}{2}(\pi - x)$, in the interval $(0, 2\pi)$.	CO4	L4										
Q-8	Find half range Cosine series for $f(x) = k$, $0 < x < 2$.	CO4	L5										
Q-9	Find the least square straight line approximation to the data <table><tr><td>x</td><td>-2</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>y</td><td>6</td><td>3</td><td>2</td><td>2</td></tr></table>	x	-2	-1	0	1	y	6	3	2	2	CO5	L3
x	-2	-1	0	1									
y	6	3	2	2									
Q-10	The regression equations calculated from a given set of observations for two random variables are $x = -0.4y + 6.4$ and $y = -0.6x + 4.6$ calculate $\bar{x}$, $\bar{y}$ and r .	CO5	L4										

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 x 10 = Marks)

		CO	BL
Q-1	Solve by the method of variation of parameters $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = e^{-x}\sec^3x$	CO1	L1
Q-2	Solve by the method of separation of variables $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2(x + y)u$	CO2	L2
Q-3	Solve $\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x} + 3x^2y$	CO2	L5
Q-4	Solve in series the given differential equation $3xy_2 + 2y_1 + y = 0$	CO3	L3
Q-5	Prove that $P_n(x)$ is the coefficient of z^n in the expansion of $(1 - 2xz + z^2)^{-1/2}$ in ascending powers of z .	CO3	L4
Q 6	Find the Fourier series for $f(x) = \begin{cases} 2, & -2 \leq x \leq 0 \\ x, & 0 < x < 2 \end{cases}$	CO4	L4
Q-7	A die is thrown 8 times and it is required to find the probability that 3 will show (i) Exactly 2 times (ii) At least 7 times (iii) At least once.	CO5	L3
Q-8	Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 8x^2e^{2x}\sin 2x$	CO1	L5